The Extension–Marginalisation Adjunction: A Categorical Statement of Paper 1

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Abstract

The Observational Extension Theorem of Paper 1 has a clean categorical formulation: the extension construction and the marginalisation functor form an adjunction

$$\mathrm{Ext} \dashv \mathrm{Forget} : \mathbf{ProbSys} \rightleftharpoons \mathbf{QuerySys}$$

between a category of *query systems with compatible measures* and a category of *probability spaces with query structure*. Under the hypotheses of Paper 1 — sequential upper-directedness, realizability, and observational determination — this adjunction is an *equivalence of categories*: both the unit and counit are natural isomorphisms.

The adjunction is the mathematical form of the program's philosophical inversion. The forgetful functor Forget extracts the marginal system from a probability space; the extension functor Ext constructs the canonical probability space from marginal data. That Ext $\dashv$ Forget is an equivalence says precisely that these two operations are inverses — that the global measure is fully determined by, and fully determines, the coherent system of its marginals.

The residual assumption — σ-algebras $\mathcal{O}_i$ on the individual query outcome spaces — is identified as the honest epistemic boundary of the program: measurability is assumed only at the level of what the observer can directly report, and the global measurable structure on the realization space is derived.

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1 Motivation: relocating the assumption

The classical construction of probability assumes a σ-algebra and a measure on a sample space Ω that the observer never directly accesses. All observations are derived from this assumed global structure. The Observable Dynamics Program inverts this: it begins with what the observer can directly report (query outcomes) and derives the global structure from the coherence of those reports.

After the inversion, the assumption has not been eliminated but *relocated*:

Level	What is assumed	Status
σ-algebras $\mathcal{O}_i$ on outcome spaces O_i	Observer's reportable distinctions	Assumed (residual)
σ-algebra $\mathcal{B}_\Omega$ on realization space Ω	Global measurable structure	*Derived* from queries
Probability measure P on Ω	Global probability	*Derived* from $\{\nu_i\}$
Objectivity	Agreement of all coherent observers	*Forced* by coherence

The residual assumption — σ-algebras on each O_i — is epistemologically grounded: it encodes the observer's capacity to distinguish outcomes at query level i. It does not presuppose any global structure. The global measurable structure on Ω is the join

$$\mathcal{B}_\Omega = \bigvee_{i \in I} \sigma(\mathrm{pr}_i) = \sigma\big(\mathrm{pr}_i^{-1}(A) : i \in I,\ A \in \mathcal{O}_i\big)\,,$$

constituted by the coherence of the queries, not presupposed beneath them.

The adjunction Ext $\dashv$ Forget is the precise mathematical statement that the second and third rows of this table are theorems.

Remark 1 (The deeper question). *The residual assumption — $\mathcal{O}_i$ on each O_i — is itself a candidate for derivation. Can the σ-algebras on outcome spaces be derived from something more primitive, such as the observer's finite discriminative capacity (Boolean closure of distinguishable events)? This is Stopping Point 1 of the program and is not addressed here. If resolved, it would eliminate the residual assumption entirely, grounding the whole measurable structure in operational discriminability.*

2 The category QuerySys of query systems

Definition 1 (Objects of **QuerySys**). *An object of* **QuerySys** *is a pair (F, ν) where:*

- $F : I^{\mathrm{op}} \to \mathbf{Meas}$ *is a* query system*: a functor from a directed preorder I^{op} to* **Meas***, giving measurable spaces $\{(O_i, \mathcal{O}_i)\}_{i \in I}$ and refinement maps $\{\pi_{ij} : O_j \to O_i\}_{i \le j}$ with $\pi_{ii} = \mathrm{id}$ and $\pi_{ij} \circ \pi_{jk} = \pi_{ik}$.*
- $\nu = \{\nu_i \in \mathcal{P}O_i\}_{i \in I}$ *is a* compatible family *of probability measures: $(\pi_{ij})_\sharp \nu_j = \nu_i$ for all $i \le j$.*

Definition 2 (Morphisms of **QuerySys**). *A morphism $(F, \nu) \to (G, \mu)$ in* **QuerySys** *— where $F : I^{\mathrm{op}} \to$* **Meas** *and $G : J^{\mathrm{op}} \to$* **Meas** *— consists of:*

- *A functor $\phi : I \to J$ (a map of index sets preserving the order).*
- *For each $i \in I$, a measurable map $f_i : O'_{\phi(i)} \to O_i$ intertwining the refinements: for all $i \le i'$ in I,*

  $$f_i \circ \pi'_{\phi(i),\phi(i')} = \pi_{i,i'} \circ f_{i'},$$

  and pushing $\mu_{\phi(i)}$ forward to ν_i:

  $$(f_i)_\sharp \mu_{\phi(i)} = \nu_i.$$

Composition of morphisms is composition of functors and pointwise composition of the measurable maps. The identity morphism has $\phi = \mathrm{id}_I$ and $f_i = \mathrm{id}_{O_i}$.

Remark 2 (Cofinality: when the construction is justified)**.** *A morphism $(\phi, \{f_i\})$ always induces a measurable map $h : \varprojlim O'_j \to \varprojlim O_i$ between realization spaces (see Section 4), and h is always measure-preserving on cylinders in the image of ϕ. The construction never breaks.*

However, if ϕ is not cofinal *— i.e. if there exist $j \in J$ with no $i \in I$ satisfying $j \leq \phi(i)$ — then h may fail to be an isomorphism: parts of Ω' indexed by J-queries outside the image of ϕ are invisible to F, and the extension $\mathrm{Ext}(\phi, \{f_i\})$ may collapse or discard measure-theoretic content.*

Cofinality is therefore not a gate that filters out illegal morphisms but a condition under which the induced map is an isomorphism *— the condition that the source query system is comprehensive enough to justify the completion it produces. It is the morphism-level analogue of realizability: just as realizability says the query projections $\mathrm{pr}_i : \Omega \to O_i$ are surjective (no part of Ω escapes the queries), cofinality says the image of ϕ is eventually dominating (no part of J escapes I).*

Condition	***Level***	***Role***
Realizability	*Within a query system*	pr_i *surjective — queries cover* Ω
Cofinality	*Between query systems*	$\phi(I)$ *cofinal in* J *— source covers target*
Both		*Extension is justified, not extrapolated*

Remark 3 (Morphisms as query comparisons)**.** *A morphism $(F, \nu) \to (G, \mu)$ says: the G-query system is at least as fine as the F-query system, and the measures are compatible under this comparison. Each G-query $\phi(i)$ projects via f_i to the F-query i, with the marginal $\mu_{\phi(i)}$ pushing forward to ν_i. When ϕ is cofinal, G is genuinely finer: it sees everything F sees and more, and the extension construction reflects this faithfully.*

3 The category ProbSys of probability spaces with query structure

Definition 3 (Objects of **ProbSys**)**.** *An object of* **ProbSys** *is a triple (Ω, P, F) where:*

- *$(\Omega, \mathcal{B}_\Omega)$ is a measurable space.*
- *$P \in \mathcal{P}\Omega$ is a probability measure.*
- *$F : I^{\mathrm{op}} \to$ **Meas** is a query system as above, together with measurable projections $\{\mathrm{pr}_i : \Omega \to O_i\}_{i \in I}$ satisfying $\pi_{ij} \circ \mathrm{pr}_j = \mathrm{pr}_i$ for all $i \leq j$, and such that*

$$\mathcal{B}_\Omega = \sigma\big(\mathrm{pr}_i^{-1}(A) : i \in I,\ A \in \mathcal{O}_i\big)\,.$$

The last condition says that $\mathcal{B}_\Omega$ is exactly *the σ-algebra generated by the query projections — no more and no less. Ω carries only the measurable structure that the queries can see.*

Definition 4 (Morphisms of **ProbSys**)**.** *A morphism $(\Omega, P, F) \to (\Omega', P', G)$ in* **ProbSys** *consists of:*

- *A measurable map $h : \Omega \to \Omega'$ with $h_\sharp P = P'$.*
- *A morphism $(\phi, \{f_i\}) : (F, \mathrm{Forget}(\Omega, P, F)) \to (G, \mathrm{Forget}(\Omega', P', G))$ in* **QuerySys** *(defined below) such that for all $i \in I$:*

$$f_i \circ \mathrm{pr}'_{\phi(i)} \circ h = \mathrm{pr}_i.$$

That is, h is a measure-preserving map that is compatible with the query projections via the query morphism $(\phi, \{f_i\})$.

Remark 4 (The σ-algebra condition)**.** *The condition $\mathcal{B}_\Omega = \sigma(\mathrm{pr}_i^{-1}(A))$ is the categorical form of* observational determination*: Ω is fully described by its query projections. It rules out probability spaces that carry "hidden" measurable structure invisible to all queries. In the program, this is guaranteed by the fact that $\Omega = \varprojlim O_i$ is constructed as a subspace of the product, so its σ-algebra is by definition generated by the coordinate projections.*

4 The functors

Definition 5 (The forgetful functor Forget)**.** *The functor* Forget : **ProbSys** $\to$ **QuerySys** *is defined as follows.*

- On objects*: $\mathrm{Forget}(\Omega, P, F) = (F, \{(\mathrm{pr}_i)_\sharp P\})$. It retains the query system F and replaces the global measure P with its marginals.*
- On morphisms*: a morphism $h : (\Omega, P, F) \to (\Omega', P', G)$ maps to the underlying query morphism $(\phi, \{f_i\})$.*

Forget *is well-defined: the marginals $\{(\mathrm{pr}_i)_\sharp P\}$ form a compatible family by functoriality of pushforward, and $(\phi, \{f_i\})$ is a morphism in* **QuerySys** *by the compatibility condition in the definition of* **ProbSys***-morphisms.*

Definition 6 (The extension functor Ext)**.** *The functor* Ext : **QuerySys** $\to$ **ProbSys** *is defined as follows.*

- On objects*: $\mathrm{Ext}(F, \nu) = (\varprojlim O_i,\ P,\ F)$ where:*
  - *$\varprojlim O_i$ is the projective limit measurable space with the subspace σ-algebra from $\prod_i O_i$,*
  - *P is the unique probability measure on $\varprojlim O_i$ with $(\mathrm{pr}_i)_\sharp P = \nu_i$ for all i, constructed by the Carathéodory extension of the finitely additive content defined on cylinder sets by the compatible family ν.*

  The σ-algebra on $\varprojlim O_i$ is by construction exactly $\sigma(\mathrm{pr}_i^{-1}(A) : i \in I, A \in \mathcal{O}_i)$, so the object condition of **ProbSys** *is satisfied.*

- On morphisms*: a morphism $(\phi, \{f_i\}) : (F, \nu) \to (G, \mu)$ in* **QuerySys** *induces a measurable map $h = \mathrm{Ext}(\phi, \{f_i\}) : \varprojlim O'_j \to \varprojlim O_i$ via the* universal property *of $\varprojlim O_i$. For each $i \in I$, define $g_i : \varprojlim O'_j \to O_i$ by*

  $$g_i(\omega') = f_i\Big(\mathrm{pr}'_{\phi(i)}(\omega')\Big).$$

  The family $\{g_i\}$ is a compatible cone over the I-diagram: for $i \le i'$,

  $$\pi_{i,i'}(g_{i'}(\omega')) = \pi_{i,i'}(f_{i'}(\mathrm{pr}'_{\phi(i')}(\omega'))) = f_i(\pi'_{\phi(i),\phi(i')}(\mathrm{pr}'_{\phi(i')}(\omega'))) = f_i(\mathrm{pr}'_{\phi(i)}(\omega')) = g_i(\omega'),$$

  using the intertwining condition on $\{f_i\}$ and compatibility of $\{\mathrm{pr}'_j\}$. The universal property of $\varprojlim O_i$ then gives a unique measurable map $h : \varprojlim O'_j \to \varprojlim O_i$ with $\mathrm{pr}_i \circ h = g_i$ for all i.

  Measure-preservation $h_\sharp Q = P$ holds on cylinders: for $A \in \mathcal{O}_i$,

  $$h_\sharp Q(\mathrm{pr}_i^{-1}(A)) = Q(g_i^{-1}(A)) = Q\Big((\mathrm{pr}'_{\phi(i)})^{-1}(f_i^{-1}(A))\Big) = \mu_{\phi(i)}(f_i^{-1}(A)) = \nu_i(A) = P(\mathrm{pr}_i^{-1}(A)),$$

using $(\mathrm{pr}'_{\phi(i)})_\sharp Q = \mu_{\phi(i)}$ *and* $(f_i)_\sharp \mu_{\phi(i)} = \nu_i$. *Since cylinders generate* $\mathcal{B}_\Omega$*, the Determination Theorem gives* $h_\sharp Q = P$.

When ϕ *is cofinal,* h *is an isomorphism; in general* h *is a well-defined measure-preserving map that may not be surjective.*

Remark 5 (Existence and uniqueness in Ext)**.** *The definition of* Ext *on objects invokes the Observational Extension Theorem (Paper 1, Theorem 6.1) for existence of* P*, and the Observational Determination Theorem (Paper 1, Theorem 4.1) for uniqueness. The extension theorem requires the query system* (F, ν) *to admit common coarsenings, sequential common refinements, and realizability; the determination theorem requires only common coarsenings.*

We include these as standing hypotheses on the objects of **QuerySys** *and* **ProbSys** *respectively, so that* Ext *and* Forget *are well-defined on the full categories.*

Remark 6 (Morphisms go beyond Paper 1)**.** *Paper 1 works entirely over a* fixed *query system. It proves existence and uniqueness of the extension for a given* (F, ν) *but says nothing about maps between different query systems.*

The morphisms $(\phi, \{f_i\}) : (F, \nu) \to (G, \mu)$ *in* **QuerySys** *— and the functoriality of* Ext *on them — are new categorical infrastructure built* on top of *Paper 1's results. The proof that* Ext *is functorial (Section 4) uses the projective limit universal property together with Paper 1's Determination Theorem, but the statement itself does not appear in Paper 1.*

This means the adjunction Ext $\dashv$ Forget *is a stronger claim than Paper 1 alone: it asserts not just that each individual query system has a canonical extension, but that these extensions are* coherently organised *into a functor between categories. Verifying this coherence — that* Ext *preserves composition and identities of query morphisms — is the additional content that the categorical picture contributes.*

5 The adjunction

Theorem 1 (Extension–marginalisation adjunction)**.** *The functors* Ext : **QuerySys** $\to$ **ProbSys** *and* Forget : **ProbSys** $\to$ **QuerySys** *form an adjunction* Ext $\dashv$ Forget*, witnessed by natural transformations:*

$$\begin{aligned} \eta &: \mathrm{id}_{\mathbf{QuerySys}} \Rightarrow \mathrm{Forget} \circ \mathrm{Ext} \qquad \textit{(unit)}, \\ \varepsilon &: \mathrm{Ext} \circ \mathrm{Forget} \Rightarrow \mathrm{id}_{\mathbf{ProbSys}} \qquad \textit{(counit)}, \end{aligned}$$

satisfying the triangle identities. Under the standing hypotheses, both η *and* ε *are natural isomorphisms, so* Ext $\dashv$ Forget *is an* equivalence of categories.

Construction of the unit η. For an object $(F, \nu) \in$ **QuerySys**, define

$$\eta_{(F,\nu)} : (F, \nu) \to \mathrm{Forget}(\mathrm{Ext}(F, \nu)) = (F, \{(\mathrm{pr}_i)_\sharp P\})$$

to be the identity morphism on the query system F, together with the identity maps $f_i = \mathrm{id}_{O_i}$. This is a valid **QuerySys**-morphism because the extension P satisfies $(\mathrm{pr}_i)_\sharp P = \nu_i$ by construction (the Observational Extension Theorem), so the pushforward condition $(f_i)_\sharp \mu_{\phi(i)} = \nu_i$ holds with $\mu_{\phi(i)} = \nu_i$ and $f_i = \mathrm{id}$.

Thus $\eta_{(F,\nu)}$ is the identity morphism in **QuerySys**. In particular it is an isomorphism. Naturality is immediate since all components are identity morphisms. □

Construction of the counit ε. For an object $(\Omega, P, F) \in$ **ProbSys**, define

$$\varepsilon_{(\Omega,P,F)} : \mathrm{Ext}(\mathrm{Forget}(\Omega, P, F)) \to (\Omega, P, F).$$

We have $\mathrm{Ext}(\mathrm{Forget}(\Omega, P, F)) = \mathrm{Ext}(F, \{(\mathrm{pr}_i)_\sharp P\}) = (\varprojlim O_i, P', F)$ where P' is the Carathéodory extension of the marginals $\{(\mathrm{pr}_i)_\sharp P\}$.

The canonical map is the measurable map

$$h : \varprojlim O_i \to \Omega, \qquad (o_i)_{i \in I} \mapsto (\text{the unique } \omega \in \Omega \text{ with } \mathrm{pr}_i(\omega) = o_i \text{ for all } i),$$

induced by the universal property of the projective limit in **Meas**: the projections $\mathrm{pr}_i : \Omega \to O_i$ form a compatible cone over the diagram F, so they factor uniquely through $\varprojlim O_i$.

This map h is an isomorphism of measurable spaces: it is measurable (by the universal property), bijective (because $\mathcal{B}_\Omega = \sigma(\mathrm{pr}_i^{-1}(A))$ means Ω is already a projective limit), and its inverse is measurable for the same reason.

Measure-preservation $h_\sharp P' = P$ follows from: both P' and $h_\sharp P'$ agree with P on all cylinder sets $\mathrm{pr}_i^{-1}(A)$, and these generate $\mathcal{B}_\Omega$, so by uniqueness (Observational Determination) $h_\sharp P' = P$.

Thus $\varepsilon_{(\Omega,P,F)} = h$ is an isomorphism. Naturality follows from the universal property of projective limits. □

Triangle identities. Since both η and ε are natural isomorphisms (in fact the unit is the identity and the counit is induced by the projective limit universal property), the triangle identities

$$\varepsilon_{\mathrm{Ext}(F,\nu)} \circ \mathrm{Ext}(\eta_{(F,\nu)}) = \mathrm{id}_{\mathrm{Ext}(F,\nu)}, \qquad \mathrm{Forget}(\varepsilon_{(\Omega,P,F)}) \circ \eta_{\mathrm{Forget}(\Omega,P,F)} = \mathrm{id}_{\mathrm{Forget}(\Omega,P,F)}$$

follow directly: the first because Ext applied to the identity morphism $\eta_{(F,\nu)}$ is the identity on $\mathrm{Ext}(F, \nu)$, and ε on the result is the identity isomorphism; the second because Forget of an isomorphism is an isomorphism and composing with the identity unit gives the identity. □

Theorem 2 (Equivalence of categories)**.** *Let* $\mathbf{QuerySys}^{\mathrm{cof}}$ *denote the subcategory of* **QuerySys** *consisting of all objects but only* cofinal *morphisms. Under the standing hypotheses (query systems sequentially upper-directed and realizable; probability spaces observationally determined),* $\mathbf{QuerySys}^{\mathrm{cof}}$ *and* **ProbSys** *are equivalent categories:*

$$\mathbf{QuerySys}^{\mathrm{cof}} \simeq \mathbf{ProbSys}.$$

The equivalence says: a query system with compatible measures and a probability space with query structure carry exactly the same information, provided query comparisons are comprehensive enough to justify the completions they produce. *Passing from one to the other via* Ext *or* Forget *loses nothing.*

On the full category **QuerySys** *(allowing non-cofinal morphisms),* Ext *is still a functor and* Ext $\dashv$ Forget *is still an adjunction, but the unit and counit are no longer isomorphisms: non-cofinal morphisms produce completions that discard information, and the adjunction witnesses this asymmetry rather than an equivalence.*

6 What the equivalence means

Remark 7 (The philosophical payoff)**.** *The equivalence* **QuerySys** $\simeq$ **ProbSys** *is the mathematical form of the program's central claim. It says that the classical starting point (a probability space with a measure) and the program's starting point (a query system with compatible marginals) are interchangeable under the program's hypotheses.*

This does not mean the two starting points are identical — Ext *and* Forget *are non-trivial constructions. It means that* no information is lost or gained in either direction*: the global measure is fully determined by the marginal system, and the marginal system is fully determined by the global measure.*

Objectivity — the global measure P — is not assumed beneath the observations but forced by their coherence*. The equivalence is the theorem that makes this precise.*

Remark 8 (What the hypotheses are doing)**.** *Paper 1 uses two distinct directedness conditions, each doing separate work. The equivalence requires all four of the following:*

1. Common coarsenings *(lower-directedness): any two queries have a common coarsening. This makes the cylinder family a π-system, which is the condition for the Determination Theorem (uniqueness of P given its marginals) and for μ_0 to be well-defined as a finitely additive content. It is needed for* Forget *to be faithful.*

2. Sequential common refinements *(sequential upper-directedness): every countable family of queries has a common refinement. This is strictly stronger than pairwise upper-directedness and is the analytic condition that forces σ-subadditivity of μ_0, allowing Carathéodory's theorem to apply. It is needed for* Ext *to be well-defined on objects.*

3. Realizability*: the projective limit $\Omega = \varprojlim O_i$ has surjective projections pr_i. This is the condition that makes σ-additivity hold: vanishing marginals imply vanishing cylinder content. Without it, σ-subadditivity fails and the Carathéodory extension may not produce a probability measure.*

4. Observational determination*: $\mathcal{B}_\Omega$ is generated by the query projections. This makes* Forget *faithful — no hidden measurable structure on Ω escapes the query system.*

Conditions 1–3 are hypotheses on objects of **QuerySys**. *Condition 4 is a hypothesis on objects of* **ProbSys** *and is automatically satisfied for objects in the image of* Ext.

The separation of conditions 1 and 2 matters: uniqueness (Determination) is cheaper *than existence (Extension).* Forget *is faithful under condition 1 alone;* Ext *needs the stronger conditions 2 and 3. The adjunction is asymmetric in this sense: it is easier to forget a global measure into its marginals than to recover a global measure from marginal data.*

If condition 3 (realizability) fails, Ext *may not exist. This is the content of Conjecture 2: can realizability be replaced by a weaker or more algebraic condition?*

Remark 9 (Connection to Kolmogorov)**.** *The classical Kolmogorov extension theorem gives the same equivalence for projective systems of Polish spaces. The difference is:*

- *Kolmogorov uses* topological *hypotheses (Polish spaces, inner regularity, Prokhorov tightness) to establish σ-additivity.*

- *The program uses* realizability *— a purely measure-theoretic condition on the projective limit — to establish σ-additivity.*

The two routes are complementary. The program's route applies in settings where the spaces need not be Polish (any standard measurable space with a realizable query structure), and the Prokhorov/equicontinuity work on the `equicontinuity-cfa-conjecture` branch is probing whether the topological route can be adapted to close Conjecture 2.

Remark 10 (The Prokhorov route: a broader Ext)**.** *Paper 1 (Section 7) establishes a second, independent construction of* Ext *that does not require the marginals $\{\nu_i\}$ to be countably additive. Instead it assumes only* finitely additive *marginals satisfying a* surjective projective uniform tightness *(SPUT) condition — a compactness hypothesis on the outcome spaces that forces σ-additivity to emerge rather than be assumed.*

In categorical terms, this extends Ext *to a broader category* $\mathbf{QuerySys}^{\mathrm{fa}}$ *whose objects have only finitely additive compatible families:*

$$\mathrm{Ext}^{\mathrm{fa}} : \mathbf{QuerySys}^{\mathrm{fa}} \to \mathbf{ProbSys}.$$

The SPUT condition plays the role that realizability plays in the countably-additive route: it is the condition under which $\mathrm{Ext}^{\mathrm{fa}}$ *is well-defined. Under shared hypotheses, both routes produce the same measure P.*

This is directly relevant to Conjecture 2: the SPUT condition is a topological/compactness condition on the outcome spaces O_i, and identifying when it can be replaced by a purely algebraic condition on the query system is one path toward closing the conjecture. The equicontinuity/CFA work on the `equicontinuity-cfa-conjecture` branch is developing exactly this topological route.

7 The residual assumption and Stopping Point 1

The equivalence **QuerySys** $\simeq$ **ProbSys** takes the σ-algebras $\mathcal{O}_i$ on the outcome spaces O_i as given. These are the *residual assumption* of the program: the assumption that has been relocated from the global space Ω to the local query outcome spaces.

The relocation is epistemologically meaningful:

- The $\mathcal{O}_i$ encode the observer's capacity to distinguish outcomes at query level i — a directly operational assumption.
- The σ-algebra $\mathcal{B}_\Omega$ is not assumed but derived as the join $\bigvee_i \sigma(\mathrm{pr}_i)$.
- The measure P is not assumed but derived by extension.
- Objectivity (P is canonical, not a free choice) is a theorem.

Conjecture 1 (Stopping Point 1 — measurability from discriminability)**.** *The σ-algebras $\mathcal{O}_i$ on query outcome spaces can themselves be derived from the observer's primitive discriminative capacity: the ability to distinguish finite families of events, closed under finite Boolean operations.*

If true, the category **QuerySys** *could be replaced by a category* **QuerySys**^{0} *of query systems with only Boolean (finitely additive) event structure, and the σ-algebras $\mathcal{O}_i$ would emerge as the minimal countably additive extension supporting coherent probabilistic reasoning. The equivalence* **QuerySys**$^0 \simeq$ **ProbSys** *would then ground the entire measurable and probabilistic structure in operational discriminability alone.*

8 Summary

Object	**In QuerySys**	**In ProbSys**
Spaces	Outcome spaces O_i (assumed)	$\Omega = \varprojlim O_i$ (constructed)
σ-algebras	$\mathcal{O}_i$ on O_i (assumed); $\mathcal{B}_\Omega$ derived	$\mathcal{B}_\Omega = \sigma(\mathrm{pr}_i^{-1}(A))$ (by construct
Measures	Compatible family $\{\nu_i\}$ (given)	Global measure P (derived)
Morphisms	Query maps $(\phi, \{f_i\})$ + measure intertwining	Measure-preserving maps + qu
Equivalence condition	ϕ cofinal (covers target queries)	Observational determination ($\mathcal{B}$
Functor	**Action**	
Ext	$(F, \nu) \mapsto (\varprojlim O_i, P, F)$ via Carathéodory extension	
Forget	$(\Omega, P, F) \mapsto (F, \{(\mathrm{pr}_i)_\sharp P\})$ via marginalisation	
Natural transformation	**Content**	
Unit η	Identity: marginals of extension = original ν_i (Extension Theorem)	
Counit ε	Projective limit universal property: $\Omega \cong \varprojlim O_i$ (Determination Theorem)	

The adjunction $\mathrm{Ext} \dashv \mathrm{Forget}$ encodes both main theorems of Paper 1 in a single categorical statement. The equivalence $\mathbf{QuerySys} \simeq \mathbf{ProbSys}$ is the categorical form of the program's philosophical inversion.